TIR-Bench详细解读

评测任务

TIR-Bench 总计包含 1215 个样本，划分为 13 项相互独立且核心的任务。基准题目分为选择题与开放作答两类，分别有 665 道与 550 道。问题文本平均长度为 46.29 字符，答案文本平均长度为 3.41 字符。

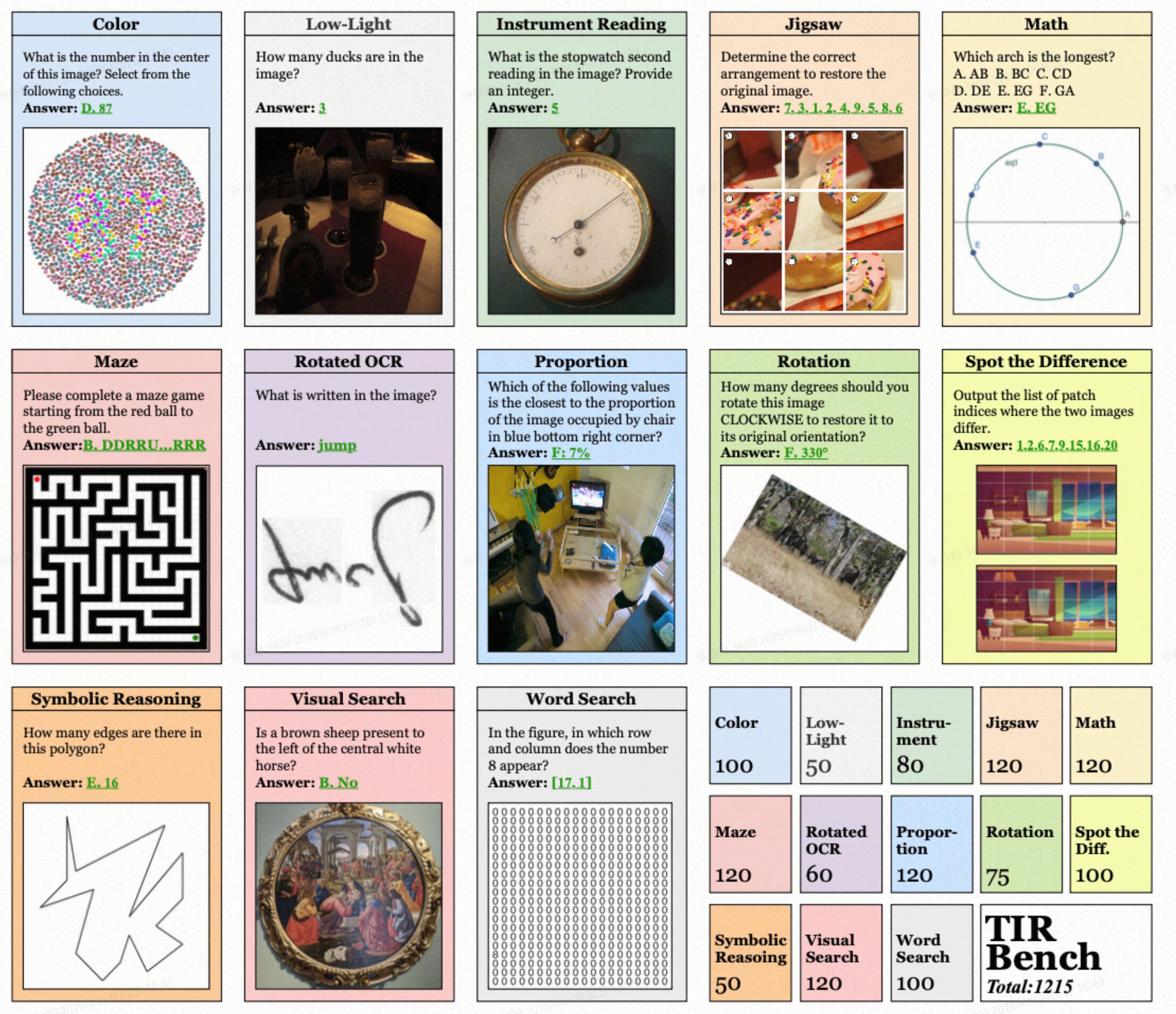

Figure 2: **Benchmark Overview.** TIR-Bench is composed of 13 tasks, meticulously designed to evaluate a wide spectrum of thinking-with-images capabilities.

评测任务介绍

任务名	中文译名	任务解释	题目原文
Color	颜色感知	评估模型对图像色彩构成相关问题的解答能力，需要模型通过编写并运行代码，程序化处理图像、计算特定颜色占比才能作答。	What is the number in the cente Select from the following choice
Low-Light	低光图像问答	评估模型在劣化视觉条件下的处理能力。面对细节被遮挡的低光图像，模型需先识别画质缺陷，再通过编写代码提升对比度、亮度，之后才能精准回答图像内容相关问题。	How many ducks are in the imag
Instrument Reading	仪器读数	评估模型多轮工具辅助分析能力。模型需先定位仪器关键读数区域，再程序化裁剪、放大特定区域以提升清晰度，最终从增强视图中精准读取数值。	What is the stopwatch second re image? Provide an integer.
Jigsaw	拼图复原	评估模型基于迭代工具流程的复杂空间推理能力。模型需程序化分割图像为若干图块，反复尝试重组，通过持续的行动与自我修正循环，评估每次组合是否复原图像，直至求解完成。	Determine the correct arrangem original image.
Math	数学几何推理	评估模型通过程序化扩充视觉输入、求解几何问题的能力。模型需使用工具绘制辅助构造线、建立坐标系，进而计算长度、比例等几何属性。	Which arch is the longest? A. AB E. EG F. GA
Maze	迷宫路径规划	评估模型高级空间规划与算法执行能力。模型需分析迷宫视觉结构，利用图像处理工具设计求解方案，应用寻路算法破解迷宫，并在图像上绘制求解路径，体现从抽象规划到具象视觉操作的完整能力。	Please complete a maze game s red ball to the green ball.
Rotated OCR	旋转图像文字识别	评估模型多步视觉推理能力。模型需先识别文本图像方向错误，再使用工具旋转至正确姿态，最后通过光学字符识别精准读取内容。	What is written in the image?
Proportion	目标占比估算	评估模型智能体能力，需要调用高性能外部分割模型获取目标掩码，再通过程序计算目标相对于整张图像的占比。	Which of the following values is proportion of the image occupie in bottom right corner?
Rotation	图像旋转矫正	评估模型迭代式方向矫正能力。要将旋转图像复原至正向姿态，模型需程序化测试各类旋转角度，每次变换后视觉评估姿态是否正确，通过工具行动与视觉判断的试错循环完成求解。	How many degrees should you r CLOCKWISE to restore it to its or orientation?
Spot the Difference	找不同	评估模型精准程序化视觉比对能力。识别两张近乎一致图像的差异时，模型需采用工具化策略，如图像像素级差分运算，高亮差异区域，再定位并标记包含差异的图像分块编号。	Output the list of patch indices v images differ.
Symbolic Reasoning	符号逻辑推理	评估模型对视觉信息应用抽象规则逻辑的能力。例如统计复杂多边形边数时，模型无法仅凭猜测作答，必须系统性识别并枚举每一条独立边，可能需要借助顶点、线条检测算法完成计数。	How many edges are there in th
Visual Search	高分辨率视觉检索	评估模型在复杂高分辨率图像中定位特定目标的多轮深度推理能力。模型需开展迭代式搜索，策略性使用工具反复放大、分析不同区域，直至定位目标或关键信息。	Is a brown sheep present to the white horse?
Word Search	字符网格单词搜索	评估模型细粒度视觉判别与异常检测能力。图像中布满高度相似字符，仅有少量细微差异，标准 OCR 在此场景下失效，迫使模型设计程序化方案，通过像素级比对、目标视觉检索等方式定位异常字符。	In the figure, in which row and c number 8 appear?

实际input数据示例：

数据

{
"meta_data": {},
"task": "refcoco",
"answer": "A",
"prompt": "Which of the following values is the closest to the proportion of the image occupied by man?\nA: 65%\nB: 73%\nC: 81%\nD: 89%\nE: 49%\nF: 57%",
"image_1": "data/refcoco/COCO_train2014_000000305267.jpg",
"image_2": null,
"id": 1
}

{
"meta_data": {},
"task": "math",
"answer": "C",
"prompt": "When x = 2.5, y \\in \nA. [2, 2.5]\nB. [1.5, 2]\nC. [2.5, 3]\nD. [3, 3.5]\nE. [3.5, 4]\nF. [1, 1.5]",
"image_1": "data/math/88.png",
"image_2": null,
"id": 5
}

{
"meta_data": {},
"task": "instrument",
"answer": "72",
"prompt": "According to the image, what is the thermometer reading in Fahrenheit? Answer as an integer like 1,2,3.",
"image_1": "data/instrument/41.jpg",
"image_2": null,
"id": 6
}

{

"meta_data": {},

"task": "word_search",

"answer": "5",

"prompt": "How many times does the letter n appear in this image? Answer with an integer, such as 1,2,3.",

"image_1": "data/word_search/38.jpg",

"image_2": null,

"id": 7

}

{

"meta_data": {

"difficulty": 3

},

"task": "jigsaw",

"answer": "[4, 1, 8, 9, 3, 6, 2, 7, 5]",

"prompt": "Instructions: Please complete the jigsaw puzzle shown in the image. The original image has been divided into {n}\\u00d7{n} pieces and scrambled. In the image, each piece is numbered from 1 to {n\\u00b2}. Your task is to determine the correct arrangement to restore the original image.\n\nQuestion: In what order should the numbered pieces be arranged to reconstruct the original image?\n\nPlease provide your answer as a sequence indicating where each numbered piece should be placed in the final arrangement (reading from left to right, top to bottom).\nFor example, 1, 16, 15, 2, 14, 8, 13, 5, 6, 7, 3, 9, 10, 11, 4, 12.",

"image_1": "data/jigsaw/shuffled_puzzle_3.jpg",

"image_2": null,

"id": 8

}

{

"meta_data": {},

"task": "maze",

"answer": "D",

"prompt": "Please complete a maze game shown in the figure. Starting from the red ball in the top-left corner, navigate the maze to reach the green ball in the bottom-right corner. 'R' means move one step to the right, 'L' means move one step to the left, 'U' means move one step up, and 'D' means move one step down. Which of the following options can successfully lead out of the maze?\nA. DDUDULRRRDDLDLRLURDLDLULUDRDDURDUDDUUDDDRULRRDLDRURDLLRURURRDRRRDDUDRULDULRU\nB. URLUULDLDLRLRDRLRULURLDLLDUULLLLRDDDDUDRRUD\nC.

LRRRRURUUULURLRUUDLULRDURRRDDRRRLRDLDDRLDDUUDDDDUDUDUDRLRURLLDULDDLRDDDDDR\nD. DDDDRRRRUURRDDRRDDDD\nE. RURULRLURLULURDLRUDRLDDRLLUDRDLLRDRUDLRDDLDURLUURUDLDDUURRRUULDRLRDDDDDLRDLLDDDRRRRRRLRRUD\nF. No answer",

"image_1": "data/maze/17.png",

"image_2": null,

"id": 10

}

{

"meta_data": {},

"task": "rotation_game",

"answer": "A",

"prompt": "How many degrees should you rotate this image CLOCKWISE to restore it to its original orientation?\n\nA. 90\\u00b0\nB. 110\\u00b0\nC. 100\\u00b0\nD. 80\\u00b0\nE. 75\\u00b0\nF. 70\\u00b0\n",

"image_1": "data/rotation_game/rotated_27.png",

"image_2": null,

"id": 4

}

{

"meta_data": {},

"task": "spot_difference",

"answer": "7, 11, 12, 13, 14, 15, 19",

"prompt": "**Task Description:**\nGiven two images of identical dimensions, each divided by thin white lines into n rows and m columns, resulting in n\\u00d7m small patches. The numbering rules are as follows:\n\n1. Patches are numbered from 1 to n\\u00d7m.\n2. Numbering follows row-major order: the first row from left to right is 1, 2, \\u2026, m; the first patch of the second row is m+1, and so on.\n\nThe i-th patch in the left image corresponds to the i-th patch in the right image. Your tasks are:\n\n1. Identify all objects or regions that differ between the two images.\n2. Return a list of patch indices occupied by each differing object.\n\n   * If an object spans multiple adjacent patches, include all those patch indices.\n3. Finally, output a deduplicated, ascending-sorted list of indices representing every patch that contains a difference.\n\n---\n\n**Output Format:**\n\n `json\n{\n  \"different_patches\": [i1, i2, i3, \\u2026]\n}\n` \n\n* `different_patches` : an array of integers containing all patch indices where the two images differ, deduplicated and sorted in ascending order.\n\n---\n\n**Example:**\nSuppose n=3 and m=4, giving 12 patches total. If patches 2, 5, 6, and 10 differ between the left and right images, the output should be:\n\n `json\n{\n  \"different_patches\": [2, 5, 6, 10]\n}\n` \n\n---\n\n**Problem:**\nOutput the list of patch indices where the two images differ.\nNote that in this image, n is 4 and m is 5",

"image_1": "data/spot_difference/79_left.png",
"image_2": "data/spot_difference/79_right.png",
"id": 11
}

{
"meta_data": {},
"task": "visual_search",
"answer": "B",
"prompt": "What is the name of the barge on the left side of the image?\nA. PETER LUND\nB. PETER LIND\nC. PETER LING\nD. PETER LEND",
"image_1": "data/visual_search/hr_9.png",
"image_2": null,
"id": 22
}

{
"meta_data": {},
"task": "color",
"answer": "C",
"prompt": "Does the horizontal bar have a uniform color? Select from the following choices. (A) Hard to tell (B) Yes (C) No",
"image_1": "data/color/109.png",
"image_2": null,
"id": 23
},

{
"meta_data": {},
"task": "ocr",
"answer": "PEAEC",
"prompt": "what is written in the image?",
"image_1": "data/ocr/47.jpg",
"image_2": null,
"id": 24
}

```json
{
  "meta_data": {},
  "task": "contrast",
  "answer": "2",
  "prompt": "How many balloons are in the image?",
  "image_1": "data/contrast/6_low.png",
  "image_2": null,
  "id": 39
}
```

```json
{
  "meta_data": {},
  "task": "symbolic",
  "answer": "E",
  "prompt": "How many edges are there in this polygon?\nA.12\nB.13\nC.14\nD.15\nE.16",
  "image_1": "data/symbolic/edge-count-4.png",
  "image_2": null,
  "id": 41
}
```

实现逻辑

阶段	对应代码入口	输入	输出	逻辑
推理	run_inference.py	TIR-Bench.json（1215 条样本）	responses.json	• 加载样本，图像转 base64 • 多线程（8 workers）并发调用 MLLM（GPT-4o、Qwen 等） • 保存模型原始回复
提取答案	extract_answer.py	responses.json	answers.json	用 few-shot demo prompt + LLM 从冗长回复中提取结构化答案
打分	calculate_score.py	answers.json	results.json + scores.txt（总分 + 分任务分数）	按任务类型使用不同评分策略 每条样本打 true_false（0/1）

论文中评测的模型

Table 2: Model Accuracy (%) Across Various Evaluation Tasks. SR: Symbolic Reasoning, WS: Word Search, LL-VQA: Low-Light VQA, IR: Instrument Reasoning, SD: Spot Difference, JG: Jigsaw Game, VS: Visual Search, RG: Rotation Game, Pro.: Proportion VQA. o3-TU: o3-tool-using, i.e., o3 with code interpreter. o3: o3 without code interpreter.

Model	All	Color	Pro.	OCR	SR	Maze	Math	WS	LL-VQA	IR	SD	JG	VS	RG
Random Guess	13.5	28.0	6.7	-	14.0	13.3	15.8	0.0	4.0	8.8	22.6	5.8	22.5	16.0
Open-Source MLLMs														
Llava-1.6-M-7B	11.3	27.0	7.5	3.3	16.0	4.2	16.7	0.0	14.0	6.3	18.0	0.0	22.5	12.0
Llava-1.6-V-7B	11.5	27.0	10.8	0.0	10.0	15.0	12.5	1.0	8.0	7.5	12.2	0.0	24.2	13.3
Llava-1.6-34B	13.0	31.0	6.7	1.7	20.0	15.8	18.3	0.0	16.0	16.3	11.9	0.0	21.7	10.7
Llava-Next-72B	11.2	20.0	15.8	3.3	8.0	10.8	15.0	0.0	10.0	11.3	16.3	0.0	23.3	12.0
Qwen2.5-VL-3B	17.7	27.0	20.0	31.7	12.0	21.7	20.8	0.0	12.0	13.8	29.7	0.0	26.7	12.0
Qwen2.5-VL-7B	16.0	21.0	10.8	48.3	14.0	15.0	24.2	0.0	22.0	11.3	24.4	0.0	21.7	9.3
Qwen2.5-VL-32B	18.7	26.0	19.2	25.0	10.0	18.3	23.3	2.0	14.0	15.0	13.1	5.4	48.3	13.3
Qwen2.5-VL-72B	19.7	37.0	15.0	33.3	24.0	35.0	22.5	3.0	32.0	12.5	14.1	0.0	25.8	12.0
InternVL3-8B	16.9	23.0	11.7	0.0	6.0	33.3	21.7	2.0	22.0	8.8	16.6	4.5	36.7	17.3
InternVL3-38B	19.1	24.0	10.8	3.3	26.0	23.3	29.2	8.0	28.0	13.8	14.6	5.1	44.2	13.3
InternVL3-78B	21.4	25.0	21.7	3.3	24.0	32.5	23.3	8.0	28.0	16.3	18.9	5.8	39.2	26.7
Proprietary MLLMs														
GPT-4.1	18.8	36.0	7.5	11.7	12.0	17.5	25.0	4.0	24.0	11.3	30.9	5.1	34.2	22.7
GPT-4o	17.3	26.0	22.5	10.0	10.0	20.0	15.8	0.0	26.0	7.5	19.4	6.2	35.0	20.0
Gemini-2.5-Flash	25.2	34.0	20.0	30.0	26.0	17.5	30.8	10.0	42.0	13.8	18.5	8.0	55.8	29.3
Gemini-2.5-Pro	28.9	44.0	21.7	25.0	34.0	24.2	30.8	12.0	42.0	20.0	28.5	10.4	**58.3**	30.7
Grok-4	22.5	35.0	**53.3**	6.7	20.0	25.8	19.2	2.00	22.0	12.5	27.0	10.0	25.8	18.7
o4-mini	21.2	39.0	17.5	8.3	12.0	13.3	21.7	5.0	30.0	18.8	33.0	8.0	39.2	26.7
o3	26.9	36.0	34.2	8.3	34.0	29.2	24.2	4.0	28.0	17.5	37.2	10.8	47.5	33.3
Tool-Using MLLMs														
DeepEyes	17.3	22.0	6.7	41.7	19.9	16.7	20.0	1.0	16.0	3.8	19.9	3.9	50.8	12.0
PyVision	31.8	53.0	26.7	**63.3**	54.0	15.8	25.8	10.0	32.0	17.5	36.4	7.6	55.0	46.7
o4-mini-TU	37.5	53.0	21.7	53.3	58.0	34.2	31.7	55.0	**44.0**	13.8	38.9	11.8	47.5	52.0
o3-TU	**46.0**	**55.0**	31.7	53.3	**66.0**	**42.5**	**50.0**	**64.0**	42.0	**21.3**	**41.0**	**16.4**	57.5	**77.3**

置信度验证

实际使用时建议先做一下验证，如论文中o3得分为26.9%，自己在本地跑一遍，如果与论文或者对应模型的技术报告中得分相符，则证明评测可靠。

qwen3.5-122b-a10b ***自测42.24% 技术报告 42.5%***	**o3** ***自测23.34% 技术报告26.9%***
answers.json all: 0.42236019073908043 color(number: 93): 0.41935483870967744 contrast(number: 50): 0.42 instrument(number: 79): 0.22784810126582278 jigsaw(number: 120): 0.15451620370370361 math(number: 120): 0.475 maze(number: 120): 0.5583333333333333 ocr(number: 60): 0.25 refcoco(number: 120): 0.5666666666666667	answers.json: 0.23341526311113736 color(number: 94): 0.3723404255319149 contrast(number: 50): 0.28 instrument(number: 80): 0.1625 jigsaw(number: 120): 0.05443055555555545 math(number: 120): 0.24166666666666667 maze(number: 120): 0.26666666666666666 ocr(number: 60): 0.06666666666666667 refcoco(number: 120): 0.35

rotation_game(number: 74): 0.2702702702702703 spot_difference(number: 100): 0.34644322344322354 symbolic(number: 50): 0.48 visual_search(number: 116): 0.7327586206896551 word_search(number: 87): 0.40229885057471265	rotation_game(number: 75): 0.3466666666666667 spot_difference(number: 100): 0.21433971171587263 symbolic(number: 50): 0.32 visual_search(number: 119): 0.35294117647058826 word_search(number: 100): 0.01

评测结论示例

结论：本次模型在 **TIR-Bench 基准上综合准确率为*%，**对照论文中的模型排名第*，第一是o3-TU 46.0%

模型静态视觉理解能力突出，其中visual_search高分辨率视觉检索（得分*%）、maze迷宫路径规划（*%）、refcoco目标占比 / 指代分割（*%），优势明显。

工具驱动的图像智能体推理能力偏弱，其中jigsaw拼图复原（*%）、ocr旋转图像文字识别（*%）、instrument仪器仪表读数（*%）、rotation_game图像旋转角度判别（*%），得分普遍偏低。

评测结论prompt：

版本一

```
你担任多模态模型评测总结助手，我会给出TIR-Bench总体准确率和各任务得分。 请生成一段精简短结论： 1. 一句话定整体水平、对标基准顶尖模型位置； 2. 简明列出核心优势任务、核心短板任务； 3. 一句话概括模型能力特点：静态视觉强弱、工具图像推理强弱； 4. 文案精炼简短，不用展开细节、不用列表。
```

版本二

```
你担任多模态大模型专业评测分析师，我会提供模型在TIR-Bench的总体准确率、各子任务名称、样本量、得分。
请按固定逻辑生成正式评测结论：
1. 开篇给出总体准确率，对标论文中 o3-TU、o4-mini-TU 做档位排名与水平定位；
2. 自动划分优势强项、中等水平、明显短板三类任务并罗列关键项；
3. 归纳模型核心能力特征：静态视觉理解表现、工具式图像智能体推理表现；
4. 分析短板根源，并给出后续迭代优化重点；
5. 行文学术正式、段落连贯，直接输出可落地到报告的结论文本，不口语、不额外解释、不列表格。
```